

# Comprehensive Model Results Summary

## Hedonic Price Index Analysis

Generated: 2026-02-28 00:28:56

14 Models: 7 Types  $\times$  2 Time Dimensions (Quarterly/Annual)

## Quarterly Models - Cumulative Jevons Indices

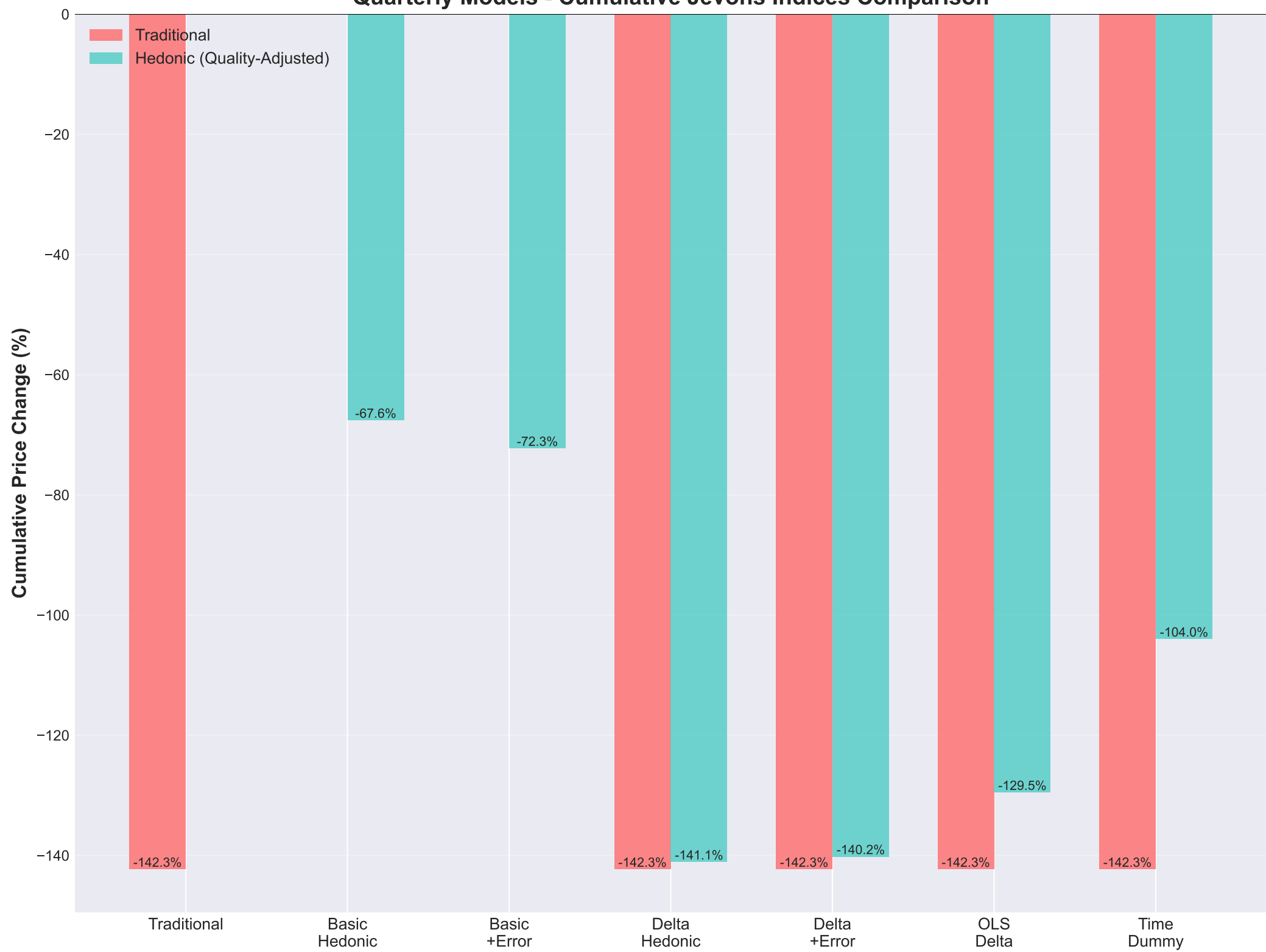
| Model                                    | Final Chained Jevons Index | Cumulative Deflation | R <sup>2</sup> (price-change)    |
|------------------------------------------|----------------------------|----------------------|----------------------------------|
| Traditional Jevons                       | -1.4231                    | -75.90%              | N/A                              |
| Levels Hedonic                           | -0.6759                    | -49.13%              | 0.1279                           |
| Levels Hedonic with Lagged Errors        | -0.7226                    | -51.45%              | 0.0936                           |
| Price-Change Hedonic                     | -1.4110                    | -75.61%              | 0.2559                           |
| Price-Change Hedonic with Lagged Errors  | -1.4020                    | -75.39%              | 0.2772                           |
| Price-Change Hedonic with OLS            | -1.2952                    | -72.62%              | 0.4364                           |
| Levels Hedonic with Time-Dummy Variables | -1.0399                    | -64.65%              | 0.0294 (partial R <sup>2</sup> ) |

# Annual Models - Cumulative Jevons Indices

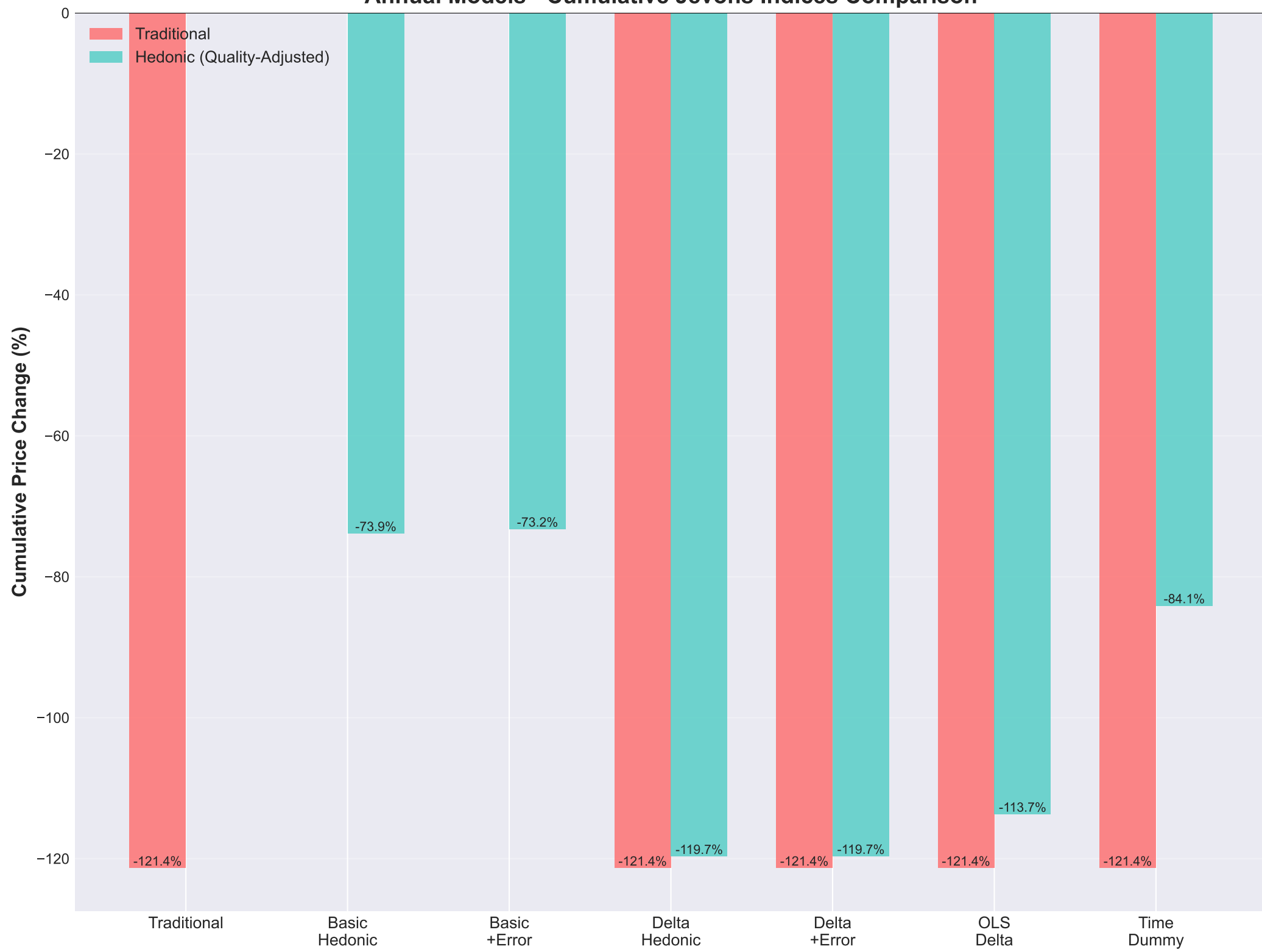
*All models cover 2020 to 2025 (5 periods)*

| Model                                    | Final Chained Jevons Index | Cumulative Deflation | R <sup>2</sup> (price-change)    |
|------------------------------------------|----------------------------|----------------------|----------------------------------|
| Traditional Jevons                       | -1.2135                    | -70.29%              | N/A                              |
| Levels Hedonic                           | -0.7387                    | -52.22%              | 0.1915                           |
| Levels Hedonic with Lagged Errors        | -0.7325                    | -51.93%              | 0.1012                           |
| Price-Change Hedonic                     | -1.1970                    | -69.79%              | 0.2498                           |
| Price-Change Hedonic with Lagged Errors  | -1.1968                    | -69.79%              | 0.2623                           |
| Price-Change Hedonic with OLS            | -1.1371                    | -67.92%              | 0.4226                           |
| Levels Hedonic with Time-Dummy Variables | -0.8412                    | -56.88%              | 0.0821 (partial R <sup>2</sup> ) |

Quarterly Models - Cumulative Jevons Indices Comparison



Annual Models - Cumulative Jevons Indices Comparison



# Model Descriptions

- **Traditional Jevons Index:**

Direct calculation using actual prices. No quality adjustment. Serves as baseline.

- **Basic Hedonic:**

Standard hedonic regression. Independent Lasso model for each period. Controls for product features.

- **Basic + Error Feature:**

Hedonic regression with previous period prediction error as additional feature. Captures time dependencies.

- **Delta Model (Lasso):**

Directly models log price differences between consecutive periods using Lasso. Uses OOF predictions to avoid mean-matching artifacts. Automatic feature selection.

- **Delta + Error Feature:**

Combines Delta model with error feature from previous period pair. Captures both price changes and time dependencies.

- **OLS Delta Model:**

Directly models log price differences using OLS (no regularization). All features included. Simpler interpretation.

- **Time Dummy Model:**

Pools consecutive periods data together. Adds time dummy variable. Single model predicts both periods. More efficient parameter sharing.